# KYLE STELLMANN

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## EDUCATION

Bachelor of Mechanical Engineering — Expected May 2028
Rowan University | Glassboro, NJ | 3.65/4.00

- President's Scholar of Excellence — Fall 2024
- Dean's List — Spring 2025 – Spring 2026

## TECHNICAL SKILLS

SolidWorks, AutoCAD, ANSYS Fluent, Cura, Arduino, Excel, CNC Machining, Lathe and Milling, MATLAB, C++

## ENGINEERING EXPERIENCE

Energy Audit Analyst | NJDMA – Rowan University — Sep 2026 – Present

- Evaluated 36 months of utility data against Heating/Cooling Degree Days (HDD/CDD) to establish a weather -normalized facility baseline and identify energy inefficiencies.
- Audited on-site HVAC systems and developed a technical Energy Conservation Measure (ECM) report, including cost-benefit analysis and calculated payback periods.

## ENGINEERING PROJECTS

Underwater Remotely Operated Vehicle (UROV)

- Executed SolidWorks Flow Simulation (CFD) on hull geometry to evaluate drag coefficients, optimizing structural contouring to reduce total drag by 15% and increase top speed by 0.3 m/s.
- Developed closed-form dynamic models balancing 10N motor thrust against hydrodynamic resistive force equations to calculate theoretical terminal velocity limits.

Precision Load Cell Scale & Embedded System

- Precision-machined an aluminum load flexure via lathe and mill to +/- 0.005" tolerances, preventing plastic deformation under 0-5 kg maximum operational loads.
- Programmed an Arduino Nano and HX711 ADC in C++ to implement digital noise filtering and sensor calibration, integrating PCB power electronics inside a water-resistant CAD enclosure.

Stirling Engine Design & Prototyping

- Modeled complete multi-body kinematic assembly in SolidWorks, applying dynamic geometric constraints and calculating flywheel rotational inertia for mechanical phase timing.
- Fabricated PLA/PVC prototypes to evaluate press-fit tolerances, thermal expansion, and low-friction sealing.

## PROFESSIONAL ORGANIZATIONS

Martinson Honors College | Member
American Society of Mechanical Engineers | Member
Assistive Technologies | Member

## WORK EXPERIENCE

Customer Service Associate | Liscio's Bakery — Sep 2025 – Present

- Delivered high-volume customer service, enforced quality standards, and optimized store workflow in a fast-paced retail environment.